SIMATS ENGINEERING (SAVEETHA SCHOOL OF ENGINEERING)

Department of Artificial Intelligence and Machine Learning

ASSESSMENT 3 – Technical Presentation(5 QUESTIONS)

Course Code:	ITA05	**Course Title:**	**Computer Vision**
CO Assessed:	CO1	**Assessment:**	Assessment 3: Technical Presentation (5 Questions × 10 Marks)
Weightage:	35% of CIA	**Total Marks:**	
CO1 Statement:	***Apply the fundamental concepts, imag models, and image processing techniq and solve real-world Computer Vision***		
Student Name:	**S.Venkateswarlu**	**Reg. No.:**	**192425340**
Section / Batch:	**2024**	**Date:**	**21/10/2026**

Code of Conduct

I S.Venkateswarlu/192425340 (NAME/ REG NO) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: S.Venkateswarlu

QUESTIONS: 5 **Total Marks: 50**

Annexure A

Technical Presentation

Q1. A shopping mall uses CCTV cameras that produce clear images during the day but dark and noisy images at night, making it difficult to identify people and objects.
Prepare a technical presentation demonstrating how filtering, noise removal, contrast enhancement, and sharpening techniques can be applied to improve the captured images. Present the sequence of image processing operations required before object detection.

Q2. A university is developing an automated document-scanning system using a camera, where captured documents may contain shadows, noise, uneven illumination, tilted pages, and blurred text. **Prepare a technical presentation explaining how filtering, thresholding, contrast enhancement, edge detection, and segmentation can be applied to improve the quality of the document images. Demonstrate the role of each technique in the processing pipeline.**

Q3. A fruit-processing company wants to automatically identify damaged apples from images captured on a conveyor belt, where shadows, reflections, different backgrounds, and brightness variations are present.
Prepare a technical presentation comparing global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation for this application. Demonstrate how a suitable technique or combination of techniques can improve fruit defect detection.

Q4. An autonomous vehicle uses a front-facing camera to detect lane markings under different weather and lighting conditions, where shadows, damaged markings, vehicles, and rain may affect the captured images.
Prepare a technical presentation demonstrating how grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and region-of-interest techniques can be applied for lane detection. Explain the challenges associated with each processing stage.

Q5. A municipal corporation wants to classify waste such as plastic, paper, and metal using a camera installed above a conveyor belt. The captured images may contain overlapping objects, shadows, reflections, noise, and different object orientations.
Prepare a technical presentation demonstrating a complete image-processing pipeline using preprocessing, enhancement, segmentation, feature extraction, and classification. Explain how the proposed pipeline can be applied to handle real-world variations and improve waste classification.

Rubrics for Calculation

Criteria	Weightage	Level 4 (Exemplary)	Level 3 (Proficient)	Level 2 (Developing)	Level 1 (Beginning)

Technical Content Accuracy	30	Highly accurate, indepth, and insightful technical explanation	Technically correct content with adequate depth and clarity	Basic concepts presented with limited accuracy and depth	Content contains major technical errors; concepts poorly understood
Problem Identification	25	Problem rigorously analyzed using appropriate and advanced methods	Problem clearly defined with a logical engineering approach	Problem identified but analysis or methodology is weak	Problem statement unclear or incorrect; no logical approach
Use of Tools	20	Advanced tools, well analyzed data, and highly effective visuals	Appropriate tools and data used effectively with clear visuals	Limited use of tools and basic data interpretation	Minimal or incorrect use of tools and data; poor visuals
Communication	15	Highly engaging, confident, and audience-focused delivery	Clear, organized, and professional communication	Understandable but lacks clarity, structure, or confidence	Poor organization, unclear speech, ineffective slides
Professionalism	10	Exemplary professionalism, leadership, and ethical awareness	Professional conduct with effective team collaboration	Basic professionalism; uneven team participation	Lacks professionalism; minimal contribution or ethical awareness

Faculty In-charge Course Coordinator Program Director

Signature: Signature: Signature:

Date: Date: Date:

Q1. Night-Time CCTV Image Enhancement Using Digital Image Processing

1. Introduction

Digital image processing is used to improve the quality of images captured by cameras. In shopping malls, CCTV cameras produce clear images during the day but dark and noisy images at night. Image enhancement techniques improve visibility and help detect people and objects accurately.

2. Problem Statement

Night-time CCTV images have low brightness, poor contrast, and unwanted noise. These problems make it difficult to identify people and objects. Therefore, image processing techniques are required to improve image quality before object detection.

3. Techniques Used

Filtering: Gaussian filtering reduces random noise, while median filtering removes salt-and-pepper noise.

Noise Removal: Removes unwanted variations in pixel values and makes images clearer.

Contrast Enhancement: Histogram equalization and CLAHE improve the difference between dark and bright regions.

Gamma Correction: Increases image brightness when gamma is less than 1.

Sharpening: Unsharp masking improves the visibility of edges and fine details.

Object Detection: A trained object detection model identifies people, vehicles, and other objects in the enhanced image.

4. Image Processing Pipeline

Input CCTV Image → Grayscale Conversion (if appropriate) → Noise Removal → Contrast Enhancement → Sharpening → Object Detection → Detection Results.

5. Tools Used

- Python
- OpenCV
- NumPy
- Matplotlib
- YOLO object detection model

- **6. Advantages**
- Improves visibility in dark images.
- Reduces unwanted noise.
- Enhances object boundaries.
- Can improve object detection performance.

7. Conclusion

Image filtering, noise removal, contrast enhancement, and sharpening improve night-time CCTV images. These techniques help object detection systems identify people and objects more effectively.

Q2. Document Image Enhancement Using Digital Image Processing

1. Introduction

Automated document scanning converts paper documents into digital images using a camera. The captured images may contain shadows, noise, tilted pages, uneven illumination, and blurred text. Digital image processing helps produce clear and readable documents.

2. Problem Statement

Poor-quality document images reduce the accuracy of text recognition. Shadows, blur, and tilted pages make the document difficult to read. Therefore, preprocessing and enhancement techniques are needed before optical character recognition (OCR).

3. Techniques Used

Filtering: Gaussian and median filters reduce noise in document images.

Contrast Enhancement: Improves the visibility of text against the background.

Thresholding: Converts grayscale images into black-and-white images. Adaptive thresholding is useful when illumination varies across the page.

Edge Detection: Canny edge detection identifies document boundaries.

Deskewing and Perspective Correction: Straightens tilted pages and corrects perspective distortion.

Segmentation: Separates text lines, words, and characters for further processing.

OCR: Converts printed or handwritten image content into digital text, depending on the OCR system's capabilities.

4. Image Processing Pipeline

Image Capture → Grayscale Conversion → Noise Removal → Contrast Enhancement → Edge Detection → Perspective Correction → Thresholding → Text Segmentation → OCR.

5. Tools Used

- Python
- OpenCV
- NumPy
- Tesseract OCR

6. Advantages

- Improves document readability.
- Removes noise and reduces shadow effects.
- Corrects tilted pages.
- Improves text recognition.
- Supports digital storage and document searching.

7. Conclusion

Filtering, thresholding, contrast enhancement, edge detection, and segmentation improve document image quality. The processed images can then be passed to OCR for extracting text.

Q3. Apple Defect Detection Using Image Segmentation

1. Introduction

Fruit-processing industries use computer vision to inspect fruit quality automatically. Image segmentation separates apples from the background and helps identify bruises, dark spots, and other visible defects.

2. Problem Statement

Apple images captured on conveyor belts may contain shadows, reflections, different backgrounds, and brightness variations. These conditions make defect detection difficult. Suitable segmentation techniques are required to separate damaged regions from healthy fruit surfaces.

3. Techniques Used

Global Thresholding: Uses a single threshold value to separate foreground and background. It is simple and fast but performs poorly under uneven illumination.

Adaptive Thresholding: Calculates thresholds for local regions. It handles uneven lighting better but may incorrectly classify natural skin texture as damage.

Edge-Based Segmentation: Detects boundaries using methods such as Canny edge detection. It helps identify apple outlines and defect boundaries.

Region-Based Segmentation: Groups neighboring pixels with similar colors or textures. It can help separate damaged regions from healthy surfaces.

4. Proposed Processing Pipeline

Image Acquisition → Noise Removal → Illumination Correction → Apple Segmentation → Defect Segmentation → Feature Extraction → Healthy or Damaged Classification → Sorting.

5. Feature Extraction

- Color: Identifies unusual brown or dark regions.
- Texture: Detects differences between smooth and damaged surfaces.
- Shape: Helps identify the apple boundary.
- Area: Measures the size of a suspected defect.

6. Tools Used

- Python

- OpenCV
- NumPy
- Scikit-learn or a deep-learning framework

7. Advantages

- Automates fruit quality inspection.
- Reduces manual inspection time.
- Improves consistency in sorting.
- Supports high-speed industrial processing.

8. Conclusion

Global thresholding, adaptive thresholding, edge-based segmentation, and region-based segmentation each have different advantages. Combining illumination correction, apple segmentation, defect analysis, and classification can improve damaged-apple detection. Shadows must be distinguished from actual defects to avoid incorrect rejection.

Q4. Lane Detection Using Digital Image Processing

1. Introduction

Lane detection is an important application of computer vision in autonomous vehicles and driver assistance systems. It identifies road lane markings and helps estimate the vehicle's position on the road.

2. Problem Statement

Road images may contain shadows, faded markings, rain, reflections, and vehicles blocking lane boundaries. These challenges make lane detection difficult under different weather and lighting conditions.

3. Techniques Used

Grayscale Conversion: Converts a color image into a single intensity channel, simplifying processing.

Filtering: Gaussian filtering reduces noise while preserving important lane boundaries.

Contrast Enhancement: Improves the visibility of faint lane markings.

Edge Detection: Canny edge detection identifies possible lane boundaries.

Thresholding: Selects pixels based on brightness or color to identify likely lane markings.

Region of Interest (ROI): Selects the road area and ignores irrelevant parts of the image.

Hough Transform: Detects line segments from edge pixels. Polynomial curve fitting can be used for curved roads.

4. Lane Detection Pipeline

Camera Image → Grayscale Conversion → Noise Filtering → Contrast Enhancement → ROI Selection → Edge Detection → Thresholding → Line or Curve Detection → Lane Boundary Estimation.

5. Tools Used

- Python
- OpenCV
- NumPy
- Matplotlib

6. Challenges and Solutions

- Shadows: Use local contrast enhancement and temporal tracking.
- Faded lines: Combine edge detection with color information.
- Rain: Use robust detection methods and consecutive video frames.

- Curved roads: Apply curve fitting instead of straight-line detection.
- Vehicles blocking lanes: Track lane estimates across frames.

7. Advantages

- Helps estimate the vehicle's lane position.
- Supports lane departure warning systems.
- Improves road-scene understanding.
- Can assist autonomous driving systems.

8. Conclusion

Grayscale conversion, filtering, contrast enhancement, edge detection, thresholding, and ROI selection form a basic lane detection pipeline. Additional tracking and robust detection methods are needed for difficult road conditions.

Q5. Automated Waste Classification Using Image Processing

1. Introduction

Waste classification is an important application of computer vision in smart waste management. A camera installed above a conveyor belt can capture waste images and classify materials such as plastic, paper, and metal.

2. Problem Statement

Waste images may contain overlapping objects, shadows, reflections, noise, and different object orientations. These variations make it difficult to identify materials correctly. An image-processing pipeline can improve object separation and classification.

3. Techniques Used

Preprocessing: Resizes images, reduces noise, and standardizes image quality.

Image Enhancement: Improves contrast and reduces the effects of uneven illumination.

Segmentation: Separates waste objects from the conveyor belt and, where possible, separates overlapping items.

Feature Extraction: Extracts color, shape, texture, and edge information.

Classification: Assigns objects to categories such as plastic, paper, and metal using machine learning or deep learning.

4. Complete Processing Pipeline

Image Acquisition → Preprocessing → Image Enhancement → Segmentation → Feature Extraction → Classification → Confidence Checking → Waste Sorting.

5. Feature Extraction

- Color: Provides information about the object's appearance.
- Shape: Helps distinguish bottles, cans, and cardboard.
- Texture: Helps differentiate paper surfaces from smooth materials.
- Edges: Describe object boundaries.
- Reflectance: May provide clues about metallic surfaces, although reflections depend on lighting.

6. Classification Methods

Rule-Based Classification: Uses predefined rules based on color, shape, or texture.

Support Vector Machine (SVM): Classifies objects using extracted numerical features.

Random Forest: Uses multiple decision trees to predict material categories.

Convolutional Neural Network (CNN): Learns image features automatically from training data.

Instance Segmentation: Detects and separates individual waste objects, even when several objects appear in one image, when the model is sufficiently trained.

7. Tools Used

- Python
- OpenCV
- NumPy
- Scikit-learn
- TensorFlow or PyTorch

8. Advantages

- Reduces manual waste sorting.
- Improves recycling efficiency.
- Supports automated material separation.
- Can reduce sorting time.
- Helps improve waste management.

9. Conclusion

A complete waste classification system combines preprocessing, enhancement, segmentation, feature extraction, and classification. Proper lighting, representative training data, and robust object segmentation help the system handle real-world variations and classify waste more accurately.